

Homework 1: Data Analysis and Statistical Decision Making

April 2026

The submission deadline is Saturday, April 25 at midnight. There will be a deduction of 10% for every day of late submission, up to 3 days. Please submit an .Rmd or .R file **and** a PDF with all your answers, tables and figures. Make sure to number the questions. Pay attention to formatting of figures and tables: include titles, legible labels, and standard errors in all regression tables. You may use AI tools to assist you in coding, but you must be able to explain every line of code and every interpretation in your submission. You may be asked to do so in office hours/recitation.

Opportunity Atlas

The data we will use for this assignment come from the Opportunity Atlas (Chetty et al., 2018). You may check the paper if you want to learn more about the data, how they were constructed, etc. The data present long-run outcomes for children who grew up in each census tract, small geographic units with an average population of 4,250 people. Our main outcome of interest is generational mobility, measured by the income of children for parents who were in the 25th percentile of the income distribution (`kfr_pooled_p25`). Outcomes are measured for a cohort born between 1978-1983, measured when they are 31-37 (around 2014).

Descriptive statistics

1. Start by looking in the <https://www.opportunityatlas.org/> at one of the following cities (choose one): Los Angeles, New York, Chicago, or New Orleans. Indicate which one you chose and describe what you observe: is there clustering in average income levels? Is there little or large variation in income levels? (1 points).
2. Using the map, within the selected city, choose a census tract. When you click on a census tract, the website will give you a number: the first two digits are the state code, the next three are the county code, and the last six digits are the tract number. This code will allow you to find the tract in the data (`atlas.csv`). How does the upward mobility of children with parents at the 25th percentile compare to the average for the same group across all tracts in the city? Report the standard deviation of `kfr_pooled_p25` across tracts in the city and compare it to the standard deviation across all tracts in the state. What do you learn from these comparisons? What do you learn from these comparisons? (2 points).

Regression analysis

3. Run a linear regression with the income of children of parents at the 25th percentile as the dependent variable and the poverty rate in 1990 as the independent variable, for each city mentioned in question 1. Generate scatter plots to visualize the results and discuss them: is the association between poverty rates and upward mobility consistent across cities? Are there cities where the association is stronger or weaker? (3 points)
4. A critic argues that racial composition is an omitted variable in your regression from question 3. Create a dummy variable called `majority_non_white` equal to 1 if `share_white2000 < 0.5`, and 0 otherwise. Using the OVB formula, explain the

expected direction of the bias. Then re-run the regression controlling for majority_non_white (for one city only). Run the additional regression needed to apply the OVB formula and calculate the predicted size of the bias. Does the coefficient on poverty rate change in the direction predicted by the formula? (2 points).

5. A colleague now suggests that the association found in (3) may not be homogeneous across tracts—specifically, that living in a high-poverty neighborhood may be particularly harmful for children in predominantly non-white communities. Adjust your model from (3) to test this hypothesis, and interpret the relevant coefficients. Present the models from (3), (4) and (5) in a nicely formatted table: include a title, descriptive labels (not variable names), titles columns, include standard errors in addition to estimates, transformation of variables so that estimates in table don't include zeros, and sufficient notes so that the table is legible as a stand-alone. (3 points).
6. Do you think any of the models in the table you made can be interpreted causally? Why? (1 points).
7. The regression above assumes that the association between poverty, racial composition, and upward mobility operates at the neighborhood level. However, it is possible that what we are really picking up is variation across cities rather than within them: some cities are simply richer, less segregated, and offer better prospects for low-income children, and children who live in all tracts of such city will have higher upward mobility. If that is true, our estimates would reflect city-level factors rather than anything specific to the neighborhood.

Dale and Krueger (2002) faced an analogous problem when estimating the returns to private schooling: students who apply to selective schools differ systematically from those who do not, so a naive comparison conflates school quality with student ambition. Their solution was to include a dummy variable for each "application group"—defined by the set of schools a student applied to—so that identification came only from students who applied to the same schools but ended up at different ones.

Apply the same logic here. Use the regression from (4) **on all data points**, and include a dummy variable for each commuting zone (cz) in your regression, so that the coefficients on poverty rate and the majority non-white dummy are identified from variation across tracts within the same city, rather than variation across cities.

*hint: use factor(cz) to dummy all commuting zones in the data. It will automatically drop one category as the reference category.

- a. Why is including a dummy for each commuting zone analogous to what Dale and Krueger did? What is the "choice set" equivalent in this context? (3 points)
- b. Report the results alongside your previous model in a single table. How do the coefficients on poverty rate and the majority non-white dummy change? (2 points)

Open question

8. Intergenerational mobility is commonly discussed in the context of urban vs. rural areas, when some argue that cities present better economic opportunities for everyone, and others showing evidence for similar levels of mobility in both urban and rural settings. You have one table, one figure, and one paragraph to provide the most elegant and convincing answer to this question possible. (4 points).

Instrumental Variable

We briefly discussed in class a paper by Angrist and Evans (1998) that studies the effect of having a third child on maternal labor supply. The core challenge is that the decision to have a third child is not random — women who choose to have more children may differ systematically from those who do not in ways that also affect their labor force participation, making OLS estimates biased. Their instrument is based on the gender composition of the first two children, captured in the variables 'gender1' and 'gender2'. The data from the paper is available directly in R:

```
install.packages("AER")
```

```
library(AER)
```

```
data("Fertility2")
```

9. The key variables are: morekids (whether the mother had a third child), work (whether the mother worked in the past year), gender1 and gender2 (the gender of the first and second child).
 - a. Create a dummy variable equal to 1 if the first two children are the same gender, and 0 otherwise. Explain why this variable is a plausible instrument for morekids. (2 points).
 - b. Run the three regressions that make up 2SLS. Write each model separately rather than using ivreg(). (2 points).
*hint: use the fitted values from the first stage as the independent variable in the second stage model.

Code Understanding Sufficiency Test

Try your best to answer using your own words. You may use any tool to help yourself understand the code, but using AI generated answers is strictly forbidden, and will be considered a violation of the homework AI policy.

This section is mean to help you (and us) asses your understanding of R code. It is not intended to penalize AI use, just guarantee that you understand the basics of your own code. **This section will not be counted towards your grade, so errors here will not be**

penalized, honest mistakes here are encouraged. However, failure to complete (as a whole or partially) will be penalized.

1. List all the libraries you used in your code. **In your own words:**
 - a. Describe the library
 - b. List 15 functions from these libraries that you use in your code. Describe what each function does.

e.g. If you use something like

```
ca <- ca %>%  
  mutate(poverty_group = ifelse(meal_pct > median(meal_pct),  
                                "High Poverty", "Low Poverty"))
```

- a. List *tidyverse* as one of the packages you used. Followed by an explanation of what *tidyverse* contains/is used for.
- b. Name *mutate()* as a tidyverse function. And explain how mutate works.
- c. List *ifelse()* and *median()* as base R functions (pre-loaded, not from your libraries), and explain how they work.